Case Study

Credit Risk & Loan Default Analysis

Problem Statement

The bank's risk management team has observed a steady increase in loan defaults but lacks clear visibility into which borrower segments contribute most to these losses. Although thousands of personal loans are approved each year, management cannot confidently determine whether default risk is primarily driven by loan purpose, borrower income, credit grade, home ownership, or previous credit behavior.

Without these insights, lending decisions rely heavily on generalized credit policies rather than data-driven risk assessment. This increases the likelihood of approving high-risk applicants while rejecting creditworthy borrowers, leading to higher credit losses, reduced profitability, and inefficient capital allocation.

The objective of this analysis is to identify the key factors influencing loan defaults, evaluate whether the current loan grading system accurately reflects borrower risk, and provide actionable recommendations to improve lending decisions and portfolio performance.

Data Overview

- **Source:** Credit Risk Dataset
- **Records:** ~32,500 Loan Applications
- **Tools Used:** Excel, MySQL, Power BI

The dataset contains applicant demographics, employment history, annual income, home ownership status, loan purpose, loan amount, interest rate, credit history, previous default records, and loan repayment status.

Tool Stack

Excel (Data Cleaning) → **MySQL** (Business Analysis) → **Power BI** (Interactive Dashboard)

Business Questions

1. What is the overall loan default rate, and how does home ownership affect it?
2. Which loan purpose has the highest default rate?
3. Does borrower income influence loan default risk?
4. Does the bank's loan grading system accurately reflect borrower risk?
5. Does a previous default on file predict future loan default?
6. Are defaulted loans charged higher interest rates than non-defaulted loans?

Data Cleaning (Excel)

Performed data validation and quality checks before analysis.

Cleaning Steps

- Verified duplicate loan records.
- Removed unrealistic age values (greater than 90 years).
- Checked for missing and blank values.
- Filled missing employment length using the dataset median.
- Filled missing loan interest rates using the average interest rate for each loan grade.
- Standardized numeric data formats.
- Verified categorical values for Home Ownership, Loan Grade, and Loan Purpose.
- Created **Loan Status Label** column for better readability.

Analysis (SQL)

Key Findings

1. What is the overall default rate and how does home ownership affect it?

total_applications	total_defaults	overall_default_rate
32575	7108	21.82

Overall default rate is **21.82%**.

Renters recorded the highest default rate, while borrowers with **mortgages** showed the lowest default rate, indicating that **home ownership** is a strong indicator of borrower financial stability.

Home_Owner	Total_Applications	total_defaults	default_rate_pct	avg_interest_rate	avg_income
RENT	16442	5192	31.58	9.95	54984.85
OTHER	107	33	30.84	10.51	76387.80
MORTGAGE	13442	1690	12.57	9.97	80677.95
OWN	2584	193	7.47	9.94	57834.81

2. Which loan purpose has the highest default rate?

Debt Consolidation loans recorded the highest default rate at **28.59%**.

This suggests borrowers applying for these loans may already be experiencing financial stress before borrowing.

Loan_Purcahse	Total_Applications	total_defaults	default_rate_pct	avg_interest_rate	avg_income
DEBTCONSOLIDATION	5212	1490	28.59	9.96	66470.88
MEDICAL	6070	1621	26.71	9.95	61443.39
HOMEIMPROVEMENT	3605	941	26.10	10.00	73549.47
PERSONAL	5520	1098	19.89	9.92	66789.48
EDUCATION	6451	1111	17.22	9.96	64111.68
VENTURE	5717	847	14.82	9.99	66352.43

3. Does borrower income affect default risk?

Borrowers earning below **$30,000** recorded a default rate of **47.08%**, compared to **13.16%** for higher-income applicants.

However, loan affordability, particularly the loan-to-income ratio, appears to be a stronger predictor of repayment behavior than income alone.

Income_Band	Total_Applications	total_defaults	default_rate_pct	avg_interest_rate	avg_income
1. Low (<30k)	3668	1727	47.08	10.06	22299.27
2. Mid (30k-60k)	14011	3614	25.79	9.86	43759.54
3. High (60k-100k)	10329	1359	13.16	9.94	74875.28
4. Very High (100k+)	4567	408	8.93	10.22	148425.05

4. Does loan grade accurately reflect borrower risk?

Loan defaults consistently increased from **Grade A** to **Grade G**, confirming that the bank's credit grading model effectively captures borrower risk.

However, Grade **G** may require further review if observed default rates exceed expected pricing assumptions.

Loan_Grade	Total_Applications	total_defaults	default_rate_pct	avg_interest_rate	avg_income
G	64	63	98.44	9.33	76773.30
F	241	170	70.54	10.28	77008.73
E	964	621	64.42	10.13	70873.11
D	3626	2141	59.05	10.00	63663.68
C	6455	1339	20.74	10.06	63980.15
B	10448	1701	16.28	9.97	66339.63
A	10777	1073	9.96	9.86	66568.21

5. Does a previous default predict future default?

Applicants with a previous default on file recorded a default rate of **37.81%**, compared to only **18.40%** among borrowers with clean credit histories.

Previous default history proved to be the strongest predictor of future loan default.

Prior_Default_On_File	Total_Applications	total_defaults	default_rate_pct	avg_interest_rate	avg_income
Y	5745	2172	37.81	10.09	65590.78
N	26830	4936	18.40	9.93	65946.09

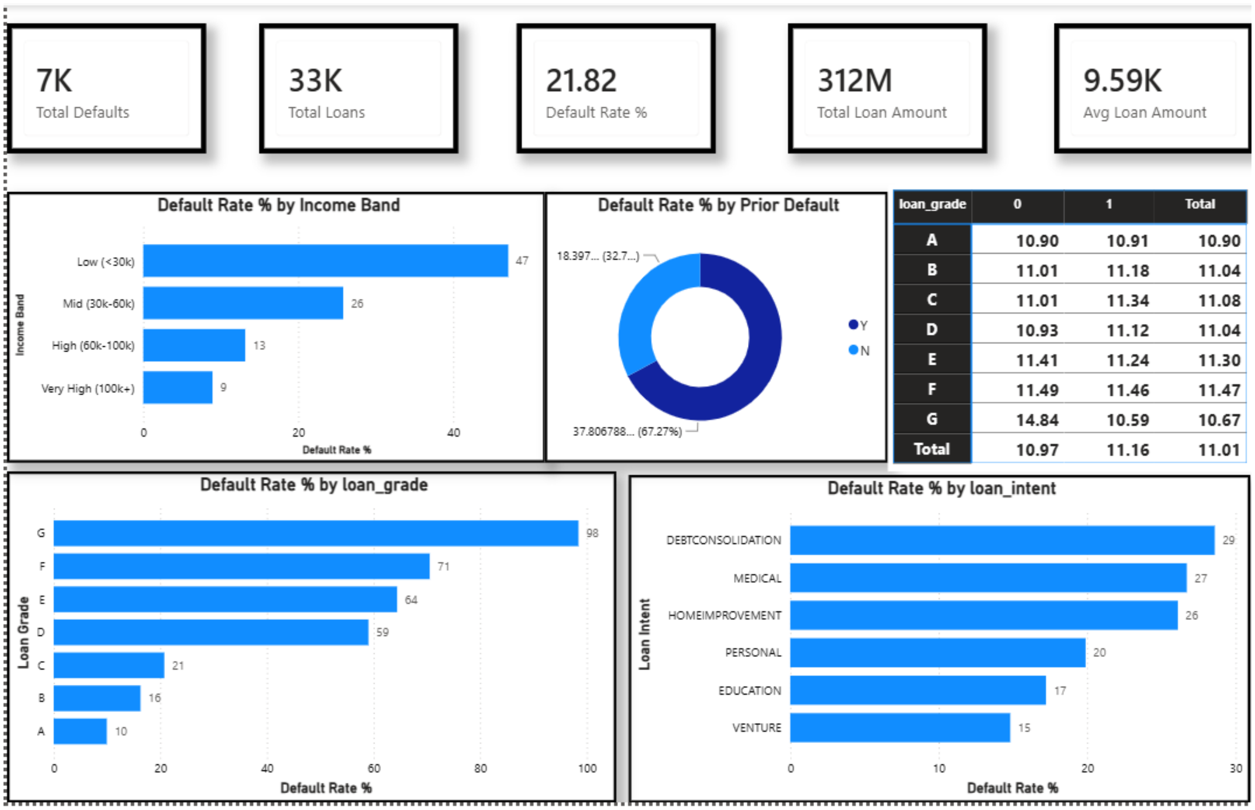
6. Are defaulted loans charged higher interest rates?

Defaulted loans carried an average interest rate of **10.05%**, compared to **9.93%** for successfully repaid loans.

Although the bank charges higher rates to riskier borrowers, current pricing may not sufficiently offset increased default losses.

loan_grade	loan_status	default_label	total_loans	avg_interest_rate	avg_loan_amount	avg_loan_to_income_ratio	avg_credit_history_years
A	0	No Default	9704	9.85	8432.11	10.90	5.8
A	1	Defaulted	1073	9.90	9508.41	10.91	5.5
B	0	No Default	8747	9.94	9751.45	11.01	5.8
B	1	Defaulted	1701	10.07	11240.70	11.16	5.6
C	0	No Default	5116	10.02	8973.27	11.02	5.9
C	1	Defaulted	1339	10.24	10141.60	11.31	5.7
D	0	No Default	1485	10.02	11137.54	10.94	6.1
D	1	Defaulted	2141	9.98	10649.28	11.12	5.8
E	0	No Default	343	10.44	13480.90	11.40	5.8
E	1	Defaulted	621	9.95	12603.74	11.25	5.8
F	0	No Default	71	10.04	14788.73	11.49	6.6
F	1	Defaulted	170	10.38	14687.50	11.46	5.9
G	0	No Default	1	14.84	1600.00	14.84	6.0
G	1	Defaulted	63	9.25	17443.25	9.25	6.5

<u>**Dashboard (Power BI)**</u>



Recommendations

1. Strengthen lending policies for repeat defaulters

Applicants with previous default history should undergo enhanced credit assessment or stricter loan approval criteria, particularly for larger loan amounts.

2. Review pricing for higher-risk loan grades

Reassess interest rate pricing for **Grade E, F, and G** borrowers to ensure expected returns adequately compensate for higher default risk.

3. Apply stricter underwriting for high-risk loan purposes

Loan purposes such as **Debt Consolidation** and **Medical** should receive additional verification and affordability checks before approval.

4. Build a composite credit risk score

Develop a lending score that combines:

- Loan Grade
- Previous Default History
- Loan-to-Income Ratio
- Employment Length

Applicants triggering multiple risk indicators should require additional approval before loan disbursement.

5. Introduce loan affordability limits

Consider limiting new loans where the loan-to-income ratio exceeds **30%**, reducing the probability of future repayment difficulties.

6. Enhance portfolio monitoring

Monitor default trends continuously across borrower segments, loan purposes, and credit grades to identify emerging risk patterns and adjust lending policies proactively.